Energy: key words and formulas

Year 8 SEAL Science · Chapter 7 Energy · One-page summary. Learn the words first, then practise the formulas.

Quantity	Formula and units	Worked example
Kinetic energy	**KE = ½ × m × v²** m in kg, v in m/s, KE in J	A 0.5 kg football kicked at 8 m/s: KE = ½ × 0.5 × 8² = ½ × 0.5 × 64 = 16 J
Gravitational potential energy	**GPE = m × g × h** g = 10 N/kg, m in kg, h in m	A 2 kg pot plant 1.5 m up: GPE = 2 × 10 × 1.5 = 30 J
Efficiency	**useful energy out ÷ total energy in × 100%** Wasted = total in − useful out	A desk fan: 70 J useful from 100 J in: 70 ÷ 100 × 100% = 70%. Wasted = 30 J
Energy used	**Energy (Wh) = power (W) × time (h)** 1 kWh = 1,000 Wh	A 750 W rice cooker for 2 h: 750 × 2 = 1,500 Wh = 1.5 kWh
Running cost	**Cost ($) = energy (kWh) × tariff ($ per kWh)** 28 cents = $0.28	1.5 kWh at $0.28 per kWh: 1.5 × 0.28 = $0.42

Convert first: grams → kilograms: ÷ 1,000 (250 g = 0.25 kg) · centimetres → metres: ÷ 100 (80 cm = 0.8 m) · watts → kilowatts: ÷ 1,000 (750 W = 0.75 kW) · minutes → hours: ÷ 60 (30 min = 0.5 h) · cents → dollars: ÷ 100 (28 c = $0.28)

Squared matters: in KE = ½mv², four times the speed gives 4² = 16 times the KE. Square the speed before you multiply.

Sankey diagram check: one input arrow on the left; useful output straight on; wasted output bent down; widths to scale; every arrow labelled with its type and amount; outputs add up to the input.

Heat transfer	What it needs	Everyday example	How to slow it
Conduction	Particles touching. Best in solids.	Your hands warm up when you hold a hot-water bottle.	Put an insulator in the way, like a cover.
Convection	A fluid (liquid or gas) that can flow.	A hot-air balloon rises because the air inside is warmer and less dense.	Stop the fluid moving, for example by closing a vent.
Radiation	Nothing. Travels as waves.	You feel the Sun's warmth on your face.	Block it with shade or a barrier.

On its own, heat flows from hotter to colder. Insulators slow heat down; they never stop it completely.

Energy basics		Efficiency and appliances	
Energy	The ability to make things happen or change. Measured in joules (J).	**Efficiency**	How much of the energy put in becomes useful energy. Never more than 100%.
Kinetic energy	Energy of movement. Thermal, light, sound and electrical energy are kinetic types.	**Sankey diagram**	An arrow diagram. Arrow widths show energy amounts, and the outputs add up to the input.
Potential energy	Stored energy. Elastic, nuclear, chemical and gravitational energy are potential types.	**Power**	How fast energy is used, in watts (W). 1 kW = 1,000 W.
Conservation of energy	Energy can't be created or destroyed. It can only be transferred or transformed.	**Kilowatt-hour (kWh)**	An amount of energy: a 1,000 W appliance running for 1 hour. 1 kWh = 1,000 Wh.
Transfer	Energy moves from one object to another. The type stays the same.	**Star rating**	A label score for how efficient an appliance is.
Transformation	Energy changes from one type to another.	**Energy consumption**	How much energy an appliance uses, in kWh.
Home design and investigations		**Tariff**	The price paid for each kWh.
Insulation and R-value	Insulation slows heat flow. A higher R-value means better insulation.		
Passive design	Designing a building to stay comfortable without appliances, for example shade and good airflow.		
Independent / dependent / controlled	What you change / what you measure / what you keep the same.		
Reliable / valid	Reliable: repeats give similar results. Valid: a fair test that measures what it claims to.		
Trend	The overall pattern in data. Describe it with numbers and units.		